

## EDUCATION

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| <b>The Ohio State University</b><br><b>M.S. Computer Science and Engineering</b>                                                                                                | Aug 2025 — May 2027<br>Columbus, OH    |
| <b>National Taiwan University of Science and Technology</b><br><b>B.S. Computer Science and Information Engineering</b><br>• Overall GPA: 3.89/4.30, Top 20% of CSIE Department | Sept 2020 — Jun 2024<br>Taipei, Taiwan |

## TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++/C#, Java, JavaScript/TypeScript, SQL
- **Frameworks & Tools:** PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV, ROS, YOLO, SLAM
- **Platforms & Technologies:** AWS, Docker, Git, REST API, Unity, Linux, CI/CD, PostgreSQL

## ENGINEERING EXPERIENCE

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| <b>Buckeye Autodrive</b> , The Ohio State University<br><b>Perception Team Member</b>                                                                                                                                                                                                                                                                                                                                                                                                                                              | Sept 2025 — Present<br>Columbus, OH     |
| <ul style="list-style-type: none"><li>• Implement perception modules for <b>autonomous driving systems</b>, leveraging <b>YOLO</b> for object detection and <b>SLAM</b> for localization within <b>ROS2</b> pipelines to enhance scene understanding.</li><li>• Build 2D/3D environment models by fusing <b>LiDAR</b> and camera data, improving obstacle detection and navigation safety.</li><li>• Collaborate with 50+ teammates across sensing, controls, and safety at OSU's CAR to align progress and integration.</li></ul> |                                         |
| <b>Artifact Tech</b> (App & Backend API Development)<br><b>Software Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Jan 2024 — Mar 2025<br>Taichung, Taiwan |
| <ul style="list-style-type: none"><li>• Designed and implemented a <b>time-series forecasting pipeline</b> for LPG demand prediction, improving accuracy by <b>10% compared to baseline averages</b> and enabling scheduled retraining.</li><li>• Built customer-facing dashboards with <b>TypeScript</b>, <b>Python</b>, and <b>SQL</b>, and integrated a <b>LINE Bot API</b> adopted by more than <b>five</b> clients, streamlining reporting workflows and enhancing user engagement.</li></ul>                                 |                                         |

## RESEARCH EXPERIENCE

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| <b>AutoMLOps-Cloud:</b> End-to-End Customer Purchase Prediction Pipeline [ <a href="#">GitHub</a> ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Remote<br>Feb 2025 — Jun 2025         |
| <b>An independent project developed while at Artifact Tech</b> <ul style="list-style-type: none"><li>• Designed and deployed a production-grade MLOps pipeline on <b>AWS Step Functions</b>, automating the full ML lifecycle from training to scheduled batch predictions.</li><li>• Containerized <b>PyTorch</b> and <b>XGBoost</b> models with <b>Docker</b> and implemented <b>CI/CD (GitHub Actions)</b> to enable reproducible deployments on <b>AWS SageMaker</b>.</li><li>• Exposed forecasts via a <b>Flask REST API</b>, reducing manual retraining effort and deployment cycle time.</li></ul> |                                       |
| <b>LERA-BFERT:</b> Live Emotional Resonance Application [ <a href="#">poster</a> ][ <a href="#">report</a> ][ <a href="#">GitHub</a> ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Taipei, Taiwan<br>Feb 2023 — Dec 2023 |
| <b>University Project led by Prof. <a href="#">Bi-Ru Da</a>, CSIE, NTUST</b> <ul style="list-style-type: none"><li>• Led a team of four to develop a <b>real-time</b> audience engagement system using dynamic facial emotion recognition and micro-expression <b>analysis</b>, improving sensitivity to subtle reactions compared to traditional FER methods.</li><li>• Conducted a pilot study with <b>five</b> participants, achieving consistent detection of micro-expressions and enabling visualization of group-level emotional trends for audience feedback analysis.</li></ul>                  |                                       |
| <b>MAE-DFER-CA:</b> Enhanced Dynamic Facial Emotion Recognition with Attention [ <a href="#">GitHub</a> ]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Taipei, Taiwan<br>Feb 2023 — Dec 2023 |
| <b>Undergraduate Research led by Prof. <a href="#">Bi-Ru Da</a>, CSIE, NTUST</b> <ul style="list-style-type: none"><li>• Implemented the existing MAE-DFER model in <b>PyTorch</b> and extended it by <b>integrating</b> the <b>CA Module (MMNET)</b>, improving <b>recognition of subtle muscle motions</b> while keeping computational cost low.</li><li>• Improved <b>model accuracy</b> to <b>WAR 52.40</b>, representing a <b>0.6% gain over 52.07</b>, while increasing FLOPS only slightly from <b>50G to 52G</b>, ensuring efficiency for real-time applications.</li></ul>                       |                                       |

## CAMPUS ACTIVITIES & AWARDS

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| <b>Volunteer, Digital Cultural Exchange Learning Project, NTUST, Taipei, Taiwan</b> [ <a href="#">post</a> ]                                                                                                                                                                                                 | Oct 2023 — Jan 2024 |
| <ul style="list-style-type: none"><li>• Mentored Kenyan students for more than twenty hours in AI software (Playground.ai) to develop ESG solutions.</li></ul>                                                                                                                                               |                     |
| <b>Team Member, E. SUN Commercial Bank, Taipei, Taiwan</b> [ <a href="#">post</a> ]                                                                                                                                                                                                                          | Oct 2023 — Dec 2023 |
| <ul style="list-style-type: none"><li>• Developed a sentiment analysis model by integrating facial expression detection and speech recognition to predict stock performance through corporate optimism.</li><li>• Won the <b>Merit Award</b> at the 2023 E.SUN BANK Business Proposal Competition.</li></ul> |                     |